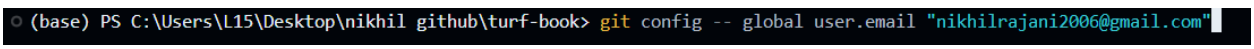
EXPERIMENT 1

AIM: TO PERFORM VERSION CONTROL OPERATIONS USING GIT
LO: This Experiment maps to LO-2

Git Commands Used

Git Command	Description
`git config --global user.email "you@example.com"`	Sets the global email address that Git associates with your commits.
`git config --global user.name "your-name"`	Sets the global username that Git associates with your commits.
`git add .`	Stages all modified and new files in the current directory for the next commit.
`git commit -m "message"`	Creates a commit with the staged changes and the given commit message.
`git push origin main`	Pushes the commits from the local main branch to the remote origin repository.
`git status`	Displays the current state of the working directory and staging area, including modified, staged, and untracked files.
`git branch`	Lists all local branches and indicates the currently active branch.
`git branch feature-branch`	Creates a new branch named feature-branch.
`git switch feature-branch`	Switches the working directory to the feature-branch branch.
`git push -u origin <branch>`	Pushes the local branch to the remote repository and sets it as the upstream tracking branch.
`git push -u origin feature-branch`	Pushes the feature-branch to the remote repository for the first time and sets it to track the remote branch.
`git commit -m "update quick sort implementation"`	Creates a commit with the staged changes and the message "update quick sort implementation".
`git push`	Pushes the latest commits from the current local branch to its corresponding remote tracking branch.
`git diff`	Shows the differences between the working directory and the staging area (unstaged changes).
`git log --oneline`	Displays the commit history in a condensed, one-line-per-commit format.
`git init`	Initializes a new Git repository (or reinitializes an existing one) in the current directory.
`git shortlog`	Summarizes the commit history grouped by author, listing each author's commit messages.
`git remote -v`	Lists the remote repositories connected to the local repo along with their fetch and push URLs.
`git fetch origin main`	Downloads commits and objects from the remote main branch without merging them into the local branch.
`git branch -a`	Lists all branches, both local and remote-tracking.
`git pull origin main`	Fetches changes from the remote main branch and merges them into the current local branch.
`git clone <repository-url>`	Creates a local copy of a remote repository, including its full history.
`git branch -M main`	Renames the current branch to main (force rename).
`git checkout -b feature-login`	Creates a new branch named feature-login and switches to it.
`git checkout feature-login`	Switches the working directory to the already-existing feature-login branch.

Terminal Output



```
(base) PS C:\Users\L15\Desktop\nikhil github\turf-book> git config -- global user.name "nikhil-rajani "
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
@nikhil-rajani ➜/workspaces/turf-book (main) $ git add .
@nikhil-rajani ➜/workspaces/turf-book (main) $ git commit -m "py added"
[main 8d9c9cf] py added
 1 file changed, 3 insertions(+)
 create mode 100644 add.py
@nikhil-rajani ➜/workspaces/turf-book (main) $ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 284 bytes | 284.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/nikhil-rajani/turf-book
   e5b0660..8d9c9cf  main -> main
@nikhil-rajani ➜/workspaces/turf-book (main) $ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
@nikhil-rajani ➜/workspaces/turf-book (main) $ git branch
* main
@nikhil-rajani ➜/workspaces/turf-book (main) $ git branch feature-branch
@nikhil-rajani ➜/workspaces/turf-book (main) $ git branch
  feature-branch
* main
@nikhil-rajani ➜/workspaces/turf-book (main) $ git switch feature-branch
Switched to branch 'feature-branch'
@nikhil-rajani ➜/workspaces/turf-book (feature-branch) $ git push -u origin branch
error: src refspec branch does not match any
error: failed to push some refs to 'https://github.com/nikhil-rajani/turf-book'
@nikhil-rajani ➜/workspaces/turf-book (feature-branch) $ git push -u origin feature-branch
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature-branch' on GitHub by visiting:
remote:      https://github.com/nikhil-rajani/turf-book/pull/new/feature-branch
remote:
To https://github.com/nikhil-rajani/turf-book
 * [new branch]      feature-branch -> feature-branch
branch 'feature-branch' set up to track 'origin/feature-branch'.
@nikhil-rajani ➜/workspaces/turf-book (feature-branch) $ git add .
@nikhil-rajani ➜/workspaces/turf-book (feature-branch) $ git commit -m "update quick sort implementation"
[feature-branch a0616ae] update quick sort implementation
 1 file changed, 2 insertions(+), 1 deletion(-)
@nikhil-rajani ➜/workspaces/turf-book (feature-branch) $ git push
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 304 bytes | 304.00 KiB/s, done.
```

```
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/nikhil-rajani/turf-book
   8d9c9cf..a0616ae  feature-branch -> feature-branch
@nikhil-rajani ➜/workspaces/turf-book (feature-branch) $ git status
On branch feature-branch
Your branch is up to date with 'origin/feature-branch'.

nothing to commit, working tree clean
```

```
@nikhil-rajani ➜/workspaces/turf-book (feature-branch) $ git diff
```

```
@nikhil-rajani ➜/workspaces/turf-book (feature-branch) $ git log --oneline
a0616ae (HEAD -> feature-branch, origin/feature-branch) update quick sort implementation
8d9c9cf (origin/main, origin/HEAD, main) py added
e5b0660 Merge branch 'main' of https://github.com/nikhil-rajani/turf-book
c897f51 initial
f095e5c Create MAIN1.html
195d03d turf
5a1f126 init
264d88a Initial commit
```

```
@nikhil-rajani ➜/workspaces/turf-book (main) $ git init
Reinitialized existing Git repository in /workspaces/turf-book/.git/
@nikhil-rajani ➜/workspaces/turf-book (main) $ git shortlog
nikhil-rajani (9):
      Initial commit
      init
      turf
      Create MAIN1.html
      initial
      Merge branch 'main' of https://github.com/nikhil-rajani/turf-book
      py added
      update quick sort implementation
      Merge pull request #1 from nikhil-rajani/feature-branch
:
nikhil-rajani (9):
      Initial commit
      init
      turf
      Create MAIN1.html
      initial
      Merge branch 'main' of https://github.com/nikhil-rajani/turf-book
      py added
      update quick sort implementation
      Merge pull request #1 from nikhil-rajani/feature-branch
:...skipping...
nikhil-rajani (9):
      Initial commit
      init
      turf
      Create MAIN1.html
      initial
      Merge branch 'main' of https://github.com/nikhil-rajani/turf-book
      py added
      update quick sort implementation
      Merge pull request #1 from nikhil-rajani/feature-branch
```

```
@nikhil-rajani ➜/workspaces/turf-book (main) $ git remote -v
origin  https://github.com/nikhil-rajani/turf-book (fetch)
origin  https://github.com/nikhil-rajani/turf-book (push)
@nikhil-rajani ➜/workspaces/turf-book (main) $ git fetch origin main
From https://github.com/nikhil-rajani/turf-book
 * branch            main       -> FETCH_HEAD
@nikhil-rajani ➜/workspaces/turf-book (main) $ git branch -a
* main
  remotes/origin/HEAD -> origin/main
  remotes/origin/feature-branch
  remotes/origin/main
@nikhil-rajani ➜/workspaces/turf-book (main) $ git pull origin main
From https://github.com/nikhil-rajani/turf-book
 * branch            main       -> FETCH_HEAD
Already up to date.
@nikhil-rajani ➜/workspaces/turf-book (main) $ git clone https://github.com/nikhil-rajani/turf-book.git
Cloning into 'turf-book'...
remote: Enumerating objects: 24, done.
remote: Counting objects: 100% (24/24), done.
remote: Compressing objects: 100% (16/16), done.
remote: Total 24 (delta 11), reused 16 (delta 4), pack-reused 0 (from 0)
Receiving objects: 100% (24/24), 15.67 KiB | 7.84 MiB/s, done.
Resolving deltas: 100% (11/11), done.
@nikhil-rajani ➜/workspaces/turf-book (main) $  git remote -v
origin  https://github.com/nikhil-rajani/turf-book (fetch)
origin  https://github.com/nikhil-rajani/turf-book (push)
@nikhil-rajani ➜/workspaces/turf-book (main) $
```

```
@nikhil-rajani ➜/workspaces/turf-book (main) $ Git branch -M main
bash: Git: command not found
@nikhil-rajani ➜/workspaces/turf-book (main) $ git checkout -b feature-login
Switched to a new branch 'feature-login'
@nikhil-rajani ➜/workspaces/turf-book (feature-login) $ git checkout feature-login
Already on 'feature-login'
@nikhil-rajani ➜/workspaces/turf-book (feature-login) $
```